

Chapter 21: Column, row, and null spaces

We're now going to connect the concept of dimension with the notion of rank.

In this section, we'll often realize vectors in $\mathbb{R}^n$ as rows instead of columns. The notions of span, linear independence, and basis are defined for rows.

Definition: Let

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

- The columns of A , considered as vectors in $\mathbb{R}^m$, span a subspace of $\mathbb{R}^m$ called the **column space** of A , denoted $\text{col}A$.
- The rows of A , considered as vectors in $\mathbb{R}^n$ (viewed as rows instead of columns), span a subspace of $\mathbb{R}^n$ called the **row space** of A , denoted $\text{row}A$.

Remark: The column space of A is exactly the same as the image of A .

Lemma: Let A and B be two $m \times n$ matrices.

- If A and B are row equivalent matrices, then the row spaces of A and B are equal, i.e. $\text{row}A = \text{row}B$.
- If A and B are column equivalent matrices, then the column spaces of A and B are equal, i.e. $\text{col}A = \text{col}B$.

Note: "Column equivalent" is the analogous definition of "row equivalent" when doing column operations instead of row operations.

Key idea in proof of (a):

A row equi. to $B \Rightarrow \exists$ elementary matrices $E_1, \dots, E_k$

s.t. $E_k \cdots E_1 \cdot A = B$. Let $E = E_k \cdots E_1$, and $E \cdot A = \begin{bmatrix} \vec{v}_1 \\ \vdots \\ \vec{v}_m \end{bmatrix} \cdot \begin{bmatrix} \vec{u}_1 \\ \vdots \\ \vec{u}_n \end{bmatrix}$

where $\vec{v}_i$'s and $\vec{u}_i$'s are the rows of E & A .

Then $\vec{v}_i \cdot A = [a_{i1}, a_{i2}, \dots, a_{in}] \cdot \begin{bmatrix} \vec{u}_1 \\ \vdots \\ \vec{u}_n \end{bmatrix} = a_{i1} \vec{u}_1 + \cdots + a_{in} \vec{u}_n = \vec{b}_i$

$\Rightarrow$ the rows of B are linear comb. of those of $A \Rightarrow \text{row}B \subseteq \text{row}A$.

Likewise, we have $A = E^{-1} \cdot B$, using the same argument, we get $\text{row}A \subseteq \text{row}B$.

$\hookrightarrow$ since E invertible

$\vec{b}_i$ is the i th row of B .

Lemma: If R is a row-echelon matrix, then

1. The nonzero rows of R are a basis of $\text{row } R$.
2. The columns of R containing leading ones are a basis of $\text{col } R$.

Proof:

1. Since they are linearly indep.
2. Other col's are linear comb. of those containing leading 1s; also the col's containing leading 1s are linearly indep.

Remark: In Chapter 2, we defined the **rank** of a matrix A to be the number of leading 1s (i.e. the number of pivots) in any row-echelon form matrix R that is row-equivalent to A . Since $\text{rank}(A) = \dim(\text{row } A)$, this justifies that the rank of A is independent of choice of R .

Example: Find a basis for $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 & 2 & -3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 & 1 & -2 \end{bmatrix}, \text{ and } \mathbf{v}_4 = \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}.$$

$$\text{Let } A = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \\ \mathbf{v}_4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix} \xrightarrow{\text{row oper.}} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

By the prev. two lemmas, $\begin{bmatrix} 1 & 1 & -1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 & -2 \end{bmatrix}$ are the basis of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$.

Theorem: Rank Theorem

Let A denote any $m \times n$ matrix of rank r . Then

$$\dim(\text{col}A) = \dim(\text{row}A) = r$$

Moreover, if A is carried to a row-echelon matrix R by row operations, then

1. The r nonzero rows of R are a basis of $\text{row}A$.
2. If the leading 1s lie in columns $j_1, j_2, \dots, j_r$ of R , then columns $j_1, j_2, \dots, j_r$ of A are a basis of $\text{col}A$. (Pick up the pivots!)

Example: Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix in the previous example.

1. What is the $\dim \text{row}A$?
2. Find a basis for the column space/image space of A .
3. What is the $\dim \text{col}A$?
4. Find a basis for the nullspace of A .

1. $\dim(\text{row}A) = 2$.

2. A basis of $\text{col}A$ consists of the col's containing leading 1s, $\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ is a basis.

3. $\dim(\text{col}A) = 2 = \dim(\text{row}A)$

4. By the prev. ex., the REF of A is $\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\Rightarrow \begin{cases} x_1 = -s \\ x_2 = 2s \\ x_3 = s \end{cases} \Rightarrow \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ is a basis of $\text{null}A$.

Definition: The **nullity** of a matrix A is the $\dim(\text{null } A)$.

Theorem: Let A denote an $m \times n$ matrix of rank r . Then

1. The $n - r$ basic solutions to the system $A\mathbf{x} = \mathbf{0}$ provided by the Gaussian algorithm are a basis of $\text{null}(A)$, so $\dim(\text{null } A) = n - r$.
2. $\dim \text{im } A = r$.
3. $\dim(\text{im } A) + \dim(\text{null } A) = n$.
4. In other words, $\text{rank}(A) + \text{nullity}(A) = n$

Fact: The following results follow from the Rank Theorem:

Let A be an $m \times n$ matrix.

- a. $\text{rank}(A) = \text{rank}(A^T)$
- b. If U is an $m \times m$ invertible matrix, and V is an $n \times n$ invertible matrix, $\text{rank}(A) = \text{rank}(UA) = \text{rank}(AV)$.

Example: Given $B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}$ and $RREF(B) = \begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$, find a basis for $(\text{im } B)$ and a basis for $\text{null } B$.

A basis for $\text{Im } B$ is $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$.

A basis for $\text{null } B$ is $\left\{ \begin{bmatrix} 1/3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \end{bmatrix} \right\}$

Example: If U is the subspace of $\mathbb{R}^3$ given by $U = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}, \begin{bmatrix} -4 \\ 6 \\ 0 \end{bmatrix} \right\}$, find

a basis for U .

$$\begin{bmatrix} -2 & 2 & -4 \\ 1 & 3 & 6 \\ -1 & 3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 1 & 3 & 6 \\ -1 & 3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ A basis of U is $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$.

Let's now add to our list of invertibility criteria:

Box Of Facts/Invertibility Criteria

Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
3. A is row equivalent to I_n .
4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
5. $\text{rank}(A) = n$.
6. A can be written as a product of elementary matrices.
7. $\det(A) \neq 0$.
8. The null space of A consists of only the zero vector.
9. The columns of A are linearly independent.
10. The columns of A form a basis for $\mathbb{R}^n$.
11. The rows of A are linearly independent.
12. The rows of A form a basis for $\mathbb{R}^n$.